

o MILESTONE 2: Core Functionalities Development – LegalEase

In this milestone, we develop the main LegalEase features: collecting document details, generating a draft through the AI model, formatting the result, displaying an editable preview, and providing document downloads.

Activity 2.1: Develop the Document Generation Function

1. Collect User Inputs

- Document type
- Parties involved
- Terms and conditions
- Effective date

2. Build a Structured Prompt

Combine the user's input into a clear prompt that asks the AI model to create a structured legal document with headings and numbered clauses.

Prompt builder program

```
def build_prompt(document_type, parties, terms, dates):  
    prompt = (  
        f"Generate a legal document titled '{document_type}'\n"  
        f"Involved parties: {parties}\n"  
        f"Effective Date: {dates}\n"  
        f"Terms and conditions: {terms}\n"  
        "Use clear headings and numbered clauses."  
    )  
    return prompt
```

3. Generate the Document

Send the prompt to the Gemini model and receive the generated text.

Activity 2.2: Format the Generated Document

1. Prepare the DOCX Output

Create a Word document with a title, section headings, paragraphs, and readable formatting.

DOCX formatter program

```
from docx import Document  
  
def format_docx(text, filename):  
    doc = Document()  
    doc.add_heading("Legal Document", 0)  
    for line in text.splitlines():  
        doc.add_paragraph(line)  
    doc.save(filename)
```

2. Prepare the PDF Output

Convert generated text into a readable PDF with appropriate spacing and page layout.

PDF formatter program

```
from fpdf import FPDF  
  
def format_pdf(text, filename):  
    pdf = FPDF()  
    pdf.add_page()
```

```
pdf.set_font("Arial", size=11)
for line in text.splitlines():
    pdf.multi_cell(0, 8, line)
pdf.output(filename)
```

3. **Prepare a Preview**

Show the generated document as readable text or styled HTML in Streamlit.

Activity 2.3: Add Editing and Download Options

1. **Allow Users to Edit the Draft**

Provide a text area where users can review and modify the generated document.

Editable preview and TXT download

```
edited_text = st.text_area(
    "Edit document",
    value=generated_text,
    height=400
)

st.download_button(
    "Download TXT",
    data=edited_text,
    file_name="legal_document.txt",
    mime="text/plain"
)
```

2. **Verify Export Formats**

- Test that TXT, DOCX, and PDF files open correctly.
- Check headings, paragraphs, and page breaks in exported documents.